Year 11 Mathematics

Shape and Space 6

10.4 Polygons

Polygons

STARTER · 5 MIN

Date: _____ **Title:** Polygons

LEARNING OBJECTIVES

- Find the sum of the interior angles of a polygon
- Find interior and exterior angles of regular polygons
- Find the number of sides of a polygon

STARTER *Answer in your book. No calculators.*

1. Work out 180 × 3
2. Work out 360 ÷ 8
3. Work out 180 − 45
4. Name a polygon with 6 sides

Starter — Answers

STARTER

Tick or correct in green pen

1. **540**
2. **45**
3. **135**
4. **Hexagon**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the sum of the interior angles of a polygon	☐
SILVER	I can find interior and exterior angles of regular polygons	☐
GOLD	I can find the number of sides of a polygon	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Angles in Polygons

INTRODUCTION · 3 MIN

A polygon with n sides can be split into n − 2 triangles.

Interior sum

(n − 2) × 180°

Pentagon: 3 × 180 = 540°

Exterior sum

Exterior angles add up to 360°.

Any polygon

Regular polygon

Exterior angle = 360 ÷ n

Hexagon: 360 ÷ 6 = 60°

Interior + exterior

They add up to 180°.

120° + 60° = 180°

Examples 1 and 2 — Interior and Exterior Angles

EXAMPLES · 5 MIN

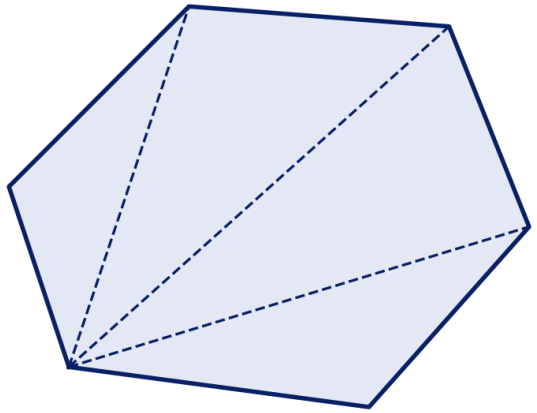
BRONZE

Watch and copy into your book

Hint interior angle sum = (n − 2) × 180°

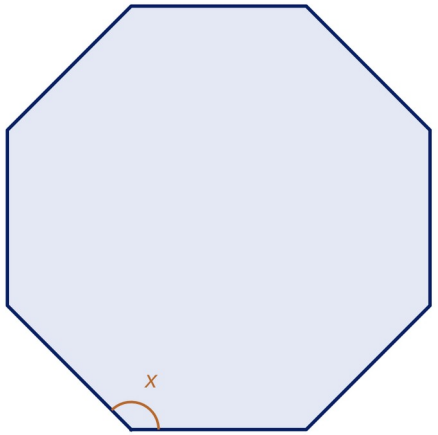
Example 1

Find the sum of the interior angles of this hexagon.



Example 2

Find x for this regular octagon.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

n = 6

(6 − 2) × 180

720°

Tip: n − 2 triangles.

Example 2

Exterior: 360 ÷ 8 = 45°

Interior: 180 − 45

135°

Tip: find the exterior angle first.

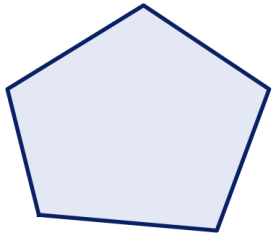
Task 1

PRACTICE · 9 MIN

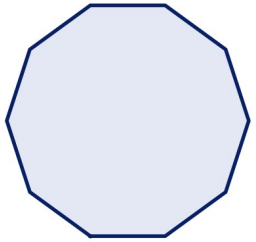
BRONZE *Answer in your book*

Hint interior angle sum = (n − 2) × 180°

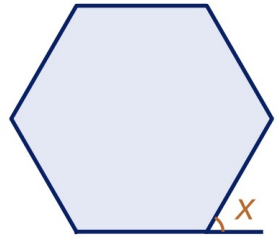
1 Sum of the interior angles?



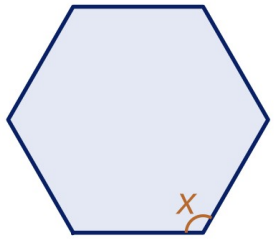
2 Sum of the interior angles?



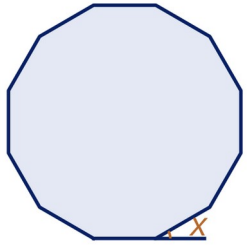
3 Regular hexagon. Find x.



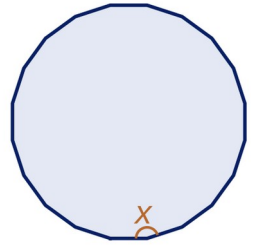
4 Regular hexagon. Find x.



5 Regular 12-sided polygon. Find x.



★ Regular 20-sided polygon. Find x.



Task 1 — Answers

CORRECTIONS

BRONZE *Tick or correct in green pen*

1. $(5-2)\times 180 = \mathbf{540°}$
2. $8\times 180 = \mathbf{1440°}$
3. $360\div 6 = \mathbf{60°}$
4. $180-60 = \mathbf{120°}$
5. $360\div 12 = \mathbf{30°}$

★ $360\div 20 = 18,\quad 180-18 = \mathbf{162°}$

Examples 3 and 4 — Missing Angles and Number of Sides

EXAMPLES · 5 MIN

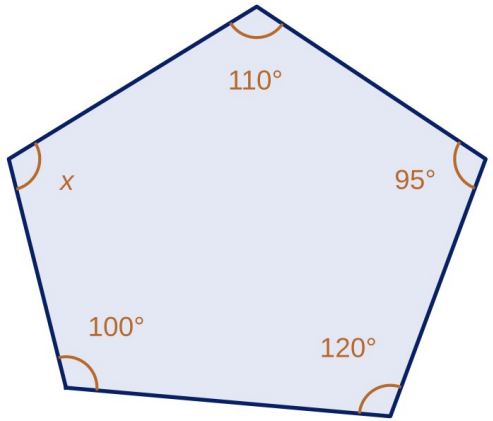
SILVER

Watch and copy into your book

Hint exterior angle = $\frac{360°}{n}$

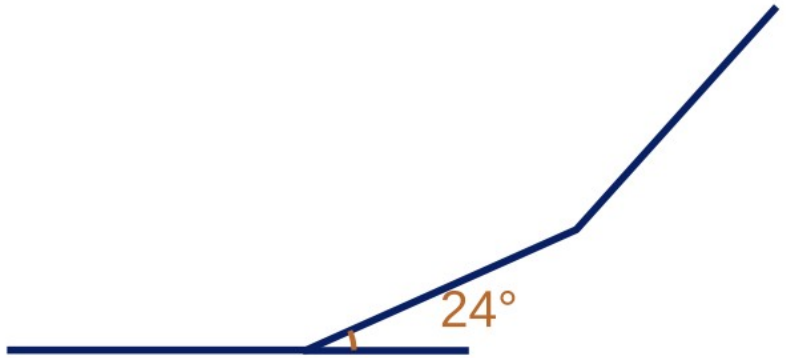
Example 3

Find x.



Example 4

Part of a regular polygon. How many sides?



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Sum = 540°

x = 540 − 425

x = 115°

Tip: find the total first.

Example 4

n = 360 ÷ 24

n = 15

Tip: 360 ÷ exterior angle = n.

Task 2

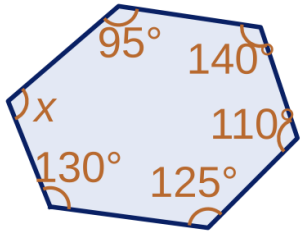
PRACTICE · 9 MIN

SILVER

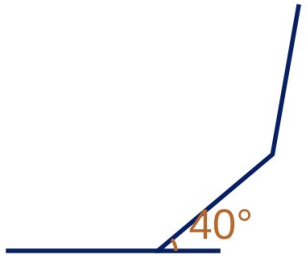
Answer in your book

Hint exterior angle = $\frac{360°}{n}$

1 Find x.



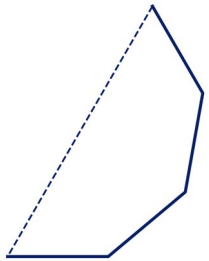
2 Part of a regular polygon. How many sides?



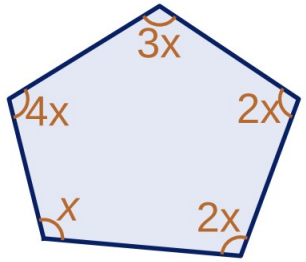
3 Part of a regular polygon. How many sides?



4 Interior angles add up to 1260°. How many sides?



5 Find x.



Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $720 - 600 = \mathbf{120°}$

2. $360 \div 40 = \mathbf{9\ sides}$

3. Exterior $30°$, $360 \div 30 = \mathbf{12\ sides}$

4. $1260 \div 180 = 7$, $\mathbf{9\ sides}$

5. $12x = 540$, $x = \mathbf{45°}$

Examples 5 and 6 — Polygon Problems

EXAMPLES · 5 MIN

GOLD

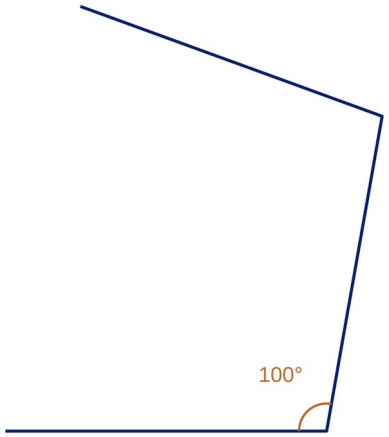
Exam-style: show all your working

Hint exterior angle = $\frac{360°}{n}$

Example 5

(3 marks)

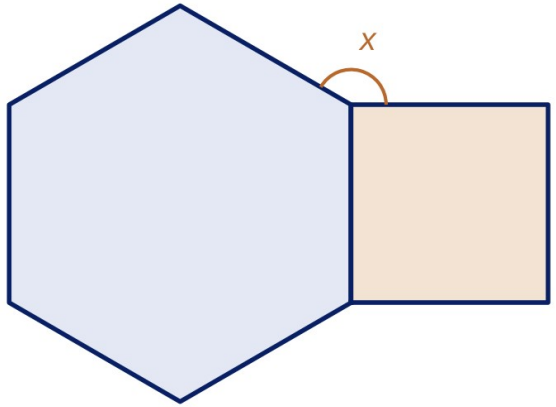
Can a regular polygon have this interior angle?



Example 6

(3 marks)

Regular hexagon and square. Find x.



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Exterior = 180 − 100

= 80°

360 ÷ 80 = 4.5

Not a whole number of sides

Tip: n must be a whole number.

Example 6

Hexagon interior = 120°

Square = 90°

360 − 120 − 90 **= 150°**

Tip: angles around a point add to 360°.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

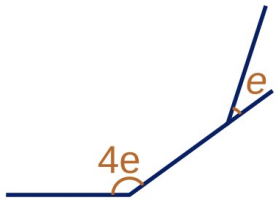
Hint exterior angle = $\frac{360°}{n}$

1 Part of a regular polygon. Is this possible?



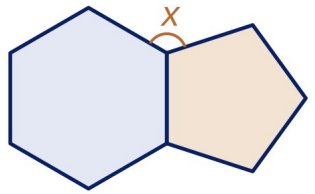
(2 marks)

2 Interior = 4 × exterior. How many sides?



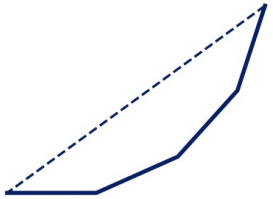
(3 marks)

3 Regular pentagon and hexagon. Find x.



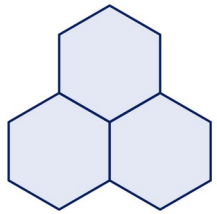
(3 marks)

4 Interior angles add up to 2340°. How many sides?



(2 marks)

5 Why do regular hexagons tessellate?



(2 marks)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $360 \div 25 = 14.4$, **No**

2. $5e = 180, \quad e = 36°$, **10 sides**

3. $108 + 120 = 228, \quad 360 - 228 =$ **132°**

4. $2340 \div 180 = 13$, **15 sides**

5. $3 \times 120 = 360$, **108 does not divide 360**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Find the sum of the interior angles of an octagon

SILVER Find the exterior angle of a regular decagon

GOLD A regular polygon has an exterior angle of 20°. How many sides?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $6 \times 180 = \mathbf{1080°}$

SILVER $360 \div 10 = \mathbf{36°}$

GOLD $360 \div 20 = \mathbf{18\ sides}$

Exam Practice — Interior and Exterior Angles

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint interior angle sum = (n − 2) × 180°

Question 1

(2 marks)

Work out the sum of the interior angles of a heptagon (7 sides).

..... °

Question 2

(3 marks)

A regular polygon has 12 sides.

(a) Work out the size of each exterior angle. (1)

..... °

(b) Work out the size of each interior angle. (2)

..... °

Exam Practice — Number of Sides

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint exterior angle = $\frac{360°}{n}$

Question 3

(2 marks)

Each exterior angle of a regular polygon is 24°.

How many sides does it have?

.....

Question 4

(3 marks)

The interior angles of a polygon add up to 1440°.

How many sides does it have?

.....

Exam Practice — Polygon Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint exterior angle = $\frac{360°}{n}$

Question 5

(2 marks)

Can a regular polygon have an interior angle of 145°? Explain.

.....

Question 6

(3 marks)

Each interior angle of a regular polygon is

4 times each exterior angle.

How many sides does it have?

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(7 − 2) × 180 — M1

= 900° — A1

Tip: a heptagon splits into 5 triangles.

Question 2

(a) 360 ÷ 12 = 30° — B1

(b) 180 − 30 — M1

= 150° — A1

Tip: interior + exterior = 180°.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

360 ÷ 24 — M1

= 15 sides — A1

Tip: n = 360 ÷ exterior angle.

Question 4

(n − 2) × 180 = 1440 — M1

n − 2 = 8 — M1

n = 10 sides — A1

Tip: divide by 180, then add 2.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

exterior = 180 − 145 = 35° — M1

360 ÷ 35 = 10.28…

No: not a whole number — A1

Tip: the number of sides must be whole.

Question 6

e + 4e = 180 — M1

e = 36° — M1

360 ÷ 36 = 10 sides — A1

Tip: interior + exterior = 180°.